Aim:
To create a basic **data analysis application** using Python in Jupyter Notebook, which takes numerical inputs and computes **mean**, **median**, and **mode** values from the dataset.

Objective:
Performing basic statistical analysis through step-by-step implementation in a Jupyter Notebook using technologies like Anaconda, Pandas, and Matplotlib.

Technologies Used

- Anaconda Distribution: Open-source Python/R distribution for scientific computing and data science.
- Jupyter Notebook: A web-based IDE for writing and executing Python code.
- Python: Programming language used for ML model development.

Step-by-Step Implementation

1. Import the Libraries:

```
import numpy as np
import pandas as pd
data = pd.read_csv("Indian_Kids_Screen_Time.csv")
data
```

2. Analyse the Mean, Median, Mode:

```
mean_value=data['Age'].mean()
print(mean_value)
```

```
12.979200988467875
```

```
mode_value=data['Age'].mode()
print(mode_value)
```

```
0    17
Name: Age, dtype: int64
```

```
median_value=data['Age'].median()
print(median_value)
```

```
13.0
```

3. Import Matplot Library:

```
import matplotlib.pyplot as plt
data['Educational_to_Recreational_Ratio'].hist()
```

4. Plot the Scatter Plot:

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8, 5))
plt.scatter(data['Age'], data['Educational_to_Recreational_Ratio'], color='purple', edgecolors='black')
plt.title("Scatter Plot of Age vs Educational_to_Recreational_Ratio")
plt.xlabel("Age")
plt.ylabel("Educational_to_Recreational_Ratio")
plt.grid(True)
plt.show()
```

5. Plot 3D Graph

```
print(data.columns.tolist())
```

```
['Age', 'Gender', 'Avg_Daily_Screen_Time_hr', 'Primary_Device', 'Exceeded_Recommended_Limit',
an_or_Rural']
```

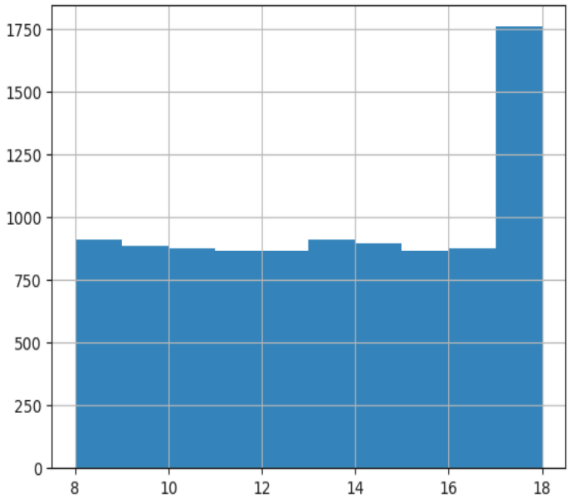
```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt

# 3D plot setup
fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

# Use valid columns from your dataset
x = data['Age']
y = data['Avg_Daily_Screen_Time_hr']
z = data['Educational_to_Recreational_Ratio']

# Plot
ax.scatter(x, y, z, c='orange', marker='o', edgecolors='black', label='3D Data Points')

# Axis labels and title
ax.set_xlabel('Age')
ax.set_ylabel('Avg Daily Screen Time (hrs)')
ax.set_zlabel('Edu to Rec Ratio')
ax.set_title('3D View: Age vs Screen Time vs Edu/Rec Ratio')

# Show legend
ax.legend()
plt.show()
```

Output:

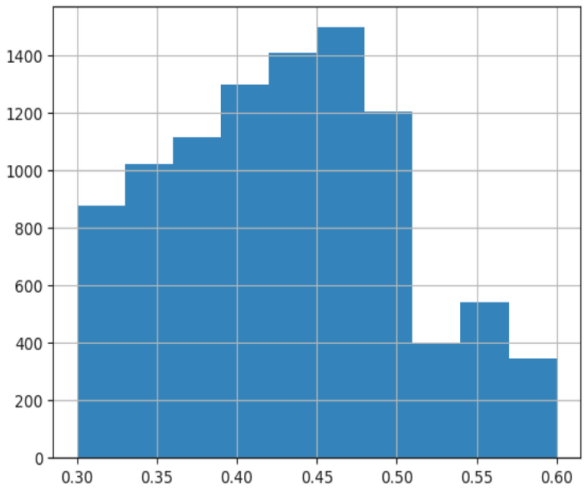
	Age	Gender	Avg_Daily_Screen_Time_hr	Primary_Device	Exceeded_Recommended_Limit	Educational_to_Recreational_Ratio	Health_Impacts	Urban_or_Rural
0	14	Male	3.99	Smartphone	True	0.42	Poor Sleep, Eye Strain	Urban
1	11	Female	4.61	Laptop	True	0.30	Poor Sleep	Urban
2	18	Female	3.73	TV	True	0.32	Poor Sleep	Urban
3	15	Female	1.21	Laptop	False	0.39	NaN	Urban
4	12	Female	5.89	Smartphone	True	0.49	Poor Sleep, Anxiety	Urban
...	...	...	...	...	...	...	...	...
9707	17	Male	3.26	Smartphone	True	0.44	Poor Sleep	Urban
9708	17	Female	4.43	Smartphone	True	0.40	Poor Sleep	Rural
9709	16	Male	5.62	Smartphone	True	0.39	Poor Sleep, Eye Strain, Anxiety	Rural
9710	17	Male	5.60	TV	True	0.43	Poor Sleep	Urban
9711	15	Female	6.12	TV	True	0.33	Anxiety	Urban

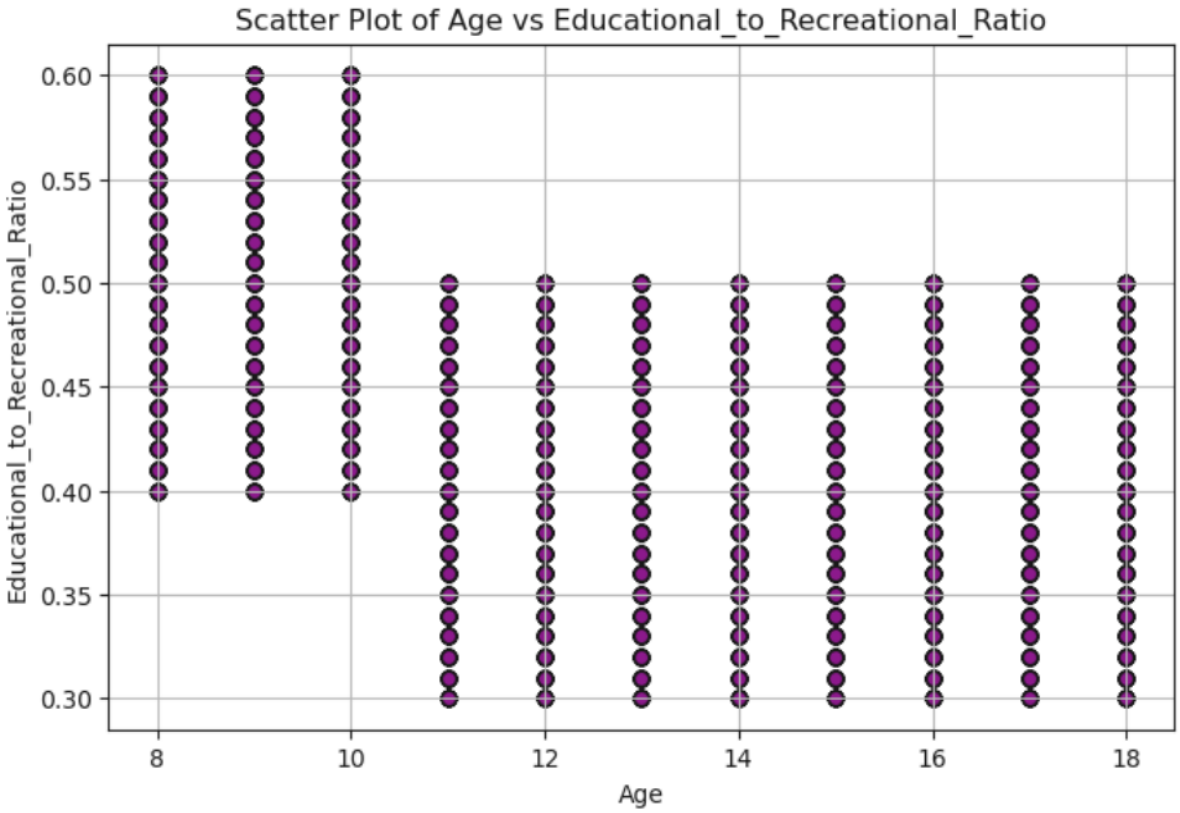
9712 rows × 8 columns

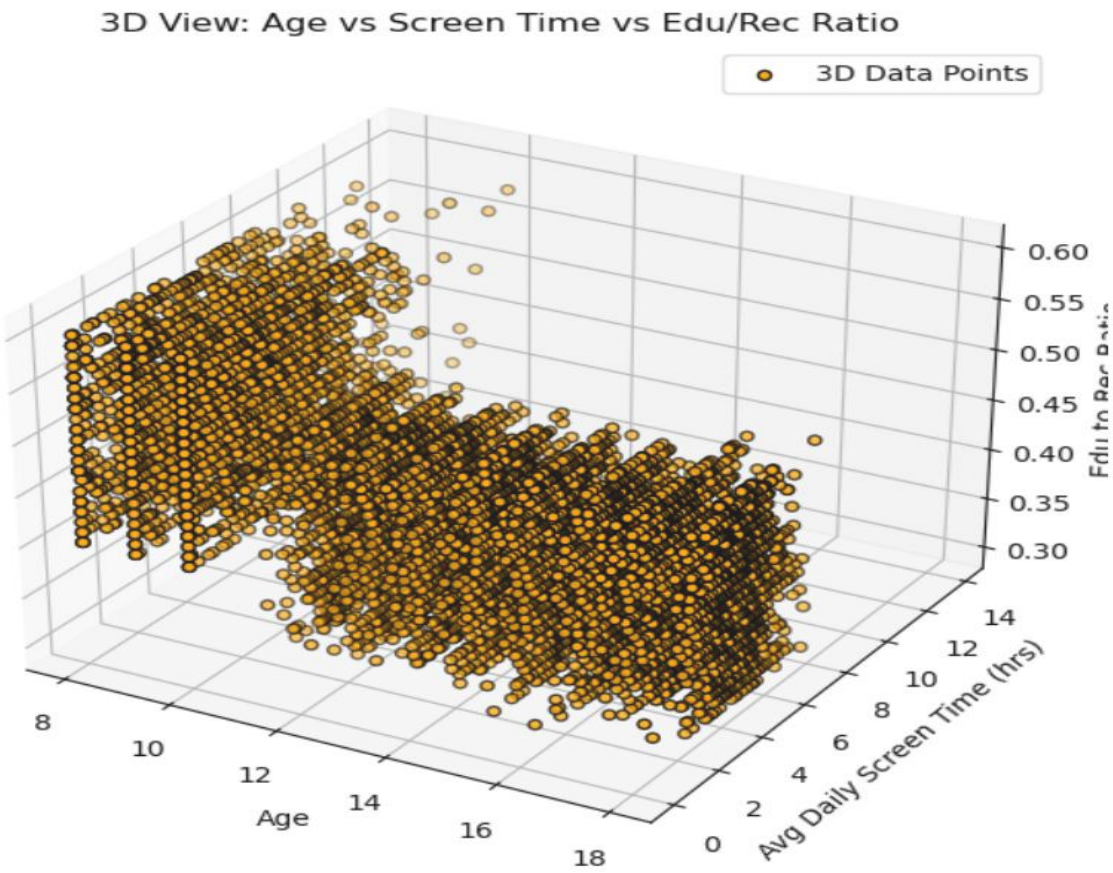
```
<Axes: >
```


1750
1500
1250
1000
750
500
250
0
8
10
12
14
16
18

```
<Axes: >
```


1400
1200
1000
800
600
400
200
0
0.30
0.35
0.40
0.45
0.50
0.55
0.60


Scatter Plot of Age vs Educational_to_Recreational_Ratio
Educational_to_Recreational_Ratio
0.60
0.55
0.50
0.45
0.40
0.35
0.30
8
10
12
14
16
18
Age



Learning Outcomes:

- Gain hands-on experience with Python tools such as Jupyter Notebook, Pandas, and Matplotlib for data analysis.
- Understand and apply basic statistical concepts like Mean, Mode, and Median on real datasets.
- Learn step-by-step implementation of statistical formulas using code in a practical environment.